

■ 1.9 Advanced Topic: Interfacing to Other Systems

■ 1.9.1 Reading Data Files

`ReadList["file", Number]` read numbers from a file, and return a *Mathematica* list of them

`ReadList["file", {Number, Number}]`
read pairs of numbers from a file

Reading numerical data files.

If you have a table of numbers in a file, generated for example by an external program, you can read them into *Mathematica* using `ReadList`. The numbers in your file can be given in C or FORTRAN-like form, as 2.3E5 and so on. `ReadList` converts all numbers to the appropriate *Mathematica* form.

This shows the contents of the file
`rand.dat`.

`In[1]:= !!rand.dat`

```
22  4.5
108 2.2e3
204 4.77e6
```

This reads the file `rand.dat`, and returns a list of the numbers in the file.

`In[1]:= ReadList["rand.dat", Number]`

`Out[1]= {22, 4.5, 108, 2200., 204, 4.77 106}`

This reads numbers from the file in pairs.

`In[2]:= ReadList["rand.dat", {Number, Number}]`

`Out[2]= {{22, 4.5}, {108, 2200.}, {204, 4.77 106}}`

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